

Dysrhythmia Secondary to Hyperkalemia



Section I: Scenario Demographics

Scenario Title:	Dysrhythmia secondary to hyperkalemia		
Date of Development:	06/04/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a situation where the diagnosis of hyperkalemia on ECG is critical.
CRM Objectives:	Lead team effectively through a case that requires administration of multiple medications.
Medical Objectives:	1) Recognize hyperkalemia as possible cause of ST elevation and treat accordingly. 2) Appropriately manage hyperkalemia with calcium, insulin and glucose, bicarbonate, ventolin, and a call for hemodialysis. 3) Recognize sine wave as indicative of hyperkalemia and alter ACLS accordingly to include administration of calcium and bicarbonate.

Case Summary: Brief Summary of Case Progression and Major Events

A 52 year-old male with end-stage renal disease (requiring dialysis) is brought in by EMS feeling weak and dizzy. He missed dialysis for the first time over the weekend to attend his niece's wedding. On presentation, his heart rate is 50 and his ECG demonstrates a wide complex rhythm with peaked T waves that EMS interprets as a STEMI. If the team recognizes the possibility of hyperkalemia and treats it appropriately, the patient's QRS will narrow. If the hyperkalemia is not recognized, the patient will arrest.

References

Pfennig CL, Slovis CM. (2013). Electrolyte disorders. In J. Marx, R. Hockberger & R. Walls (Eds.), Rosen's emergency medicine - concepts and clinical practice. pp. (1636-51). Philadelphia, PA:Saunders.

Website: accessed on April 14, 2015: <http://lifeinthefastlane.com/ecg-library/basics/hyperkalaemia/>



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
None.			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Geoff is a 52 year old male who is brought to the ED by EMS as a STEMI activation. He is not having chest pain, but has been feeling weak and dizzy today. He is diabetic and hypertensive and was started on hemodialysis 3 months ago for ESRD. He missed dialysis on the weekend for the first time so that he could attend his niece's wedding.

B. Patient Profile and History

Patient Name: Geoff Normandy	Age: 52	Weight: 80kg
Gender: ☒ M ☐ F	Code Status: Full	

Chief Complaint: Weak and dizzy

History of Presenting Illness: **Felt unwell since waking up this morning. Feels light headed with standing and like he "has no energy."** Nauseous. Missed dialysis on the weekend to attend his niece's wedding.

Past Medical History:	DM2	Medications:	Ramipril 5mg OD
	HTN		Levothyroxine 100mcg OD
	Hypothyroid		Insulin
	Dyslipidemia		Crestor 20mg OD

Allergies: Penicillin

Social History: Worked as a banker, but is now unemployed. No EtOH/smoking/illicit drug use.

Family History: No hx CAD. All have diabetes.

Review of Systems:	CNS:	Feels lightheaded, dizzy.		
	HEENT:	Nil.		
	CVS:	No CP. No palps.		
	RESP:	No SOB.		
	GI:	Nausea.		
	GU:	Anuric.		
	MSK:	Nil.	INT:	Nil.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 50/min	BP: 92/65	RR: 16/min
O ₂ SAT: 97% RA		
Rhythm: Idioventricular	T: 36.4°C	Glucose: 12.7 mmol/L
GCS: 13 (E3 V4 M6)		
General Status: Slightly diaphoretic. Looks unwell.		
CNS:	A+Ox3. No focal neurologic deficits. No asterixis.	
HEENT:	PERLA. 3mm.	
CVS:	Normal S1/S2. No murmur.	
RESP:	GAEB. No adventitious sounds.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	Nil.	SKIN: No rashes.

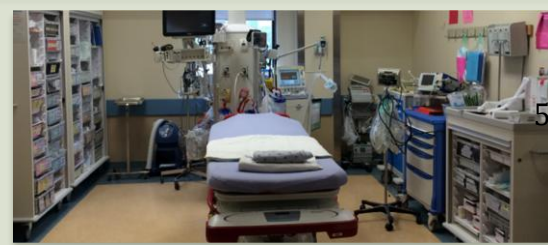
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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: idioventricular HR: 50/min BP: 92/65 RR: 20/min O ₂ SAT: 97% RA T: 36.4°C	Diaphoretic, awake, answering questions. Feels dizzy.	Learner Actions - ☐ IV, O ₂ , monitors - ☐ ECG - ☐ Blood work including trop - ☐ Fluid bolus - ☐ Check cap sugar (12.7) - ☐ Administer calcium gluconate 1g (10-30ml) iv - ☐ ± ASA 160mg PO chew	Modifiers <i>Changes to patient condition based on learner action</i> - Calcium given → HR 70, QRS narrows (change to sinus) - Fluid bolus → BP 102/67 Triggers <i>For progression to next state</i> - Calcium not given by 3 min → 2. Arrest - Initial work-up complete and calcium given → 4. Blood work back
2. Arrest Rhythm → Asystole BP → -/- RR → 0 O ₂ SAT → ?	Unresponsive with no vitals.	Learner Actions - ☐ Quality CPR - ☐ Calcium chloride 1-2 amps - ☐ NaHCO ₃ amps - ☐ Epinephrine amps - ☐ 1 amp D50 then insulin R 10 units iv	Modifiers - 7 minutes into case → critical VBG back showing hyperkalemia Triggers - 2 amps calcium given or 10 minutes into case → 3. NSR
3. NSR Rhythm → sinus HR → 70 BP → 105/70 RR → 16 O ₂ SAT → 97% RA	Awake and responding.	Learner Actions - ☐ Ventolin 20mg nebulized or 24 puffs - ☐ 1amp D50 then insulin R 10 units iv (if not yet done) - ☐ Repeat ECG - ☐ ± HCO ₃ amp - ☐ Call nephro re: dialysis	Modifiers - Ventolin given → HR 90 Triggers - Two treatments for hyperK given and Nephro called → End Case - 12 minutes → End Case
4. Blood work Back Rhythm → sinus HR → 70 BP → 105/70	Awake, alert, no longer feels dizzy.	Learner Actions - ☐ Ventolin 20mg nebulized or 24 puffs - ☐ 1amp D50 then insulin R 10 units iv - ☐ Repeat ECG - ☐ ± HCO ₃ amp - ☐ ± Lasix 40mg iv (if still makes urine) - ☐ Call nephro re: dialysis	Modifiers - Given critical VBG result showing hyperkalemia as state starts - Ventolin given → HR 90 Triggers - Two treatments for hyperK given and Nephro called → End Case - 12 minutes → End Case

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

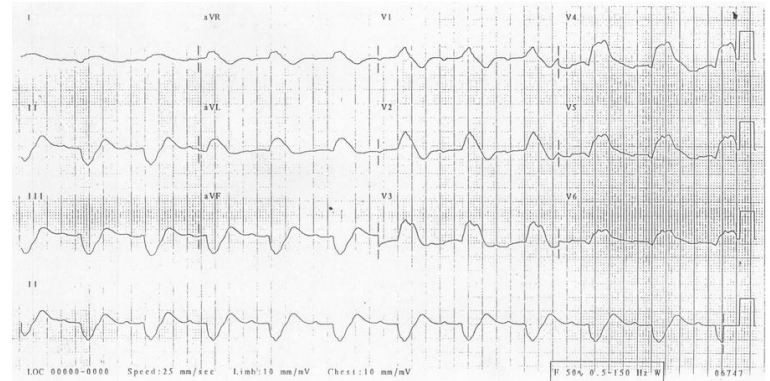
Critical VBG:

pH **7.27**
PCO2 35
PO2 45
HCO3 20
Lactate **2.5**
Na 142
K **8.6**
Cl 105

Images (ECGs, CXRs, etc.)

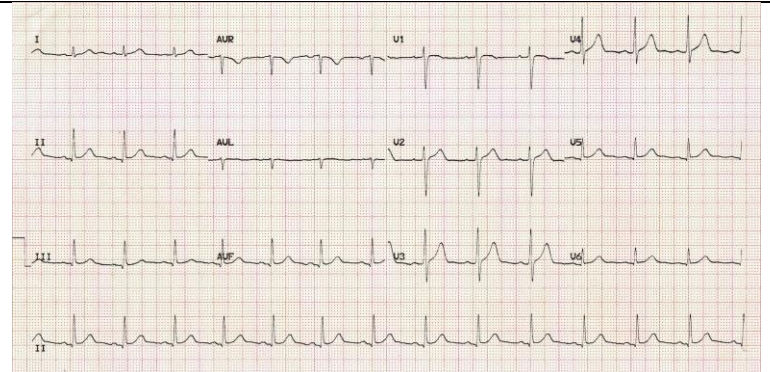
ECG 1: Hyperkalemia STEMI mimic

(Source: <http://lifeinthefastlane.com/ecg-library/basics/hyperkalaemia/>)



ECG 2: NSR with no ST changes

(Source: <http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/normal-sinus-rhythm.jpg>)



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Medical Objectives:	1) Recognize hyperkalemia as possible cause of ST elevation and treat accordingly. 2) Appropriately manage hyperkalemia with calcium, insulin, ventolin, and a call for hemodialysis. 3) Recognize sine wave as indicative of hyperkalemia and alter ACLS accordingly to include administration of calcium and bicarbonate.
Sample Questions for Debriefing	
1) What are the ECG changes associated with hyperkalemia? 2) What are some mimics of ST elevation on an ECG? 3) List 7 treatments for hyperkalemia with specific doses.	
Key Moments	
Identification of hyperkalemia on ECG.	
Alteration of ACLS management to include copious calcium and HCO ₃ in context of hyperkalemia	
Recognition of need to offer multiple treatments for hyperkalemia and of need to call nephro for dialysis.	